

Project Report: Full Stack Web Application Using MERN Stack

Project Title:

Task Manager Web Application (Full Stack – MERN)

Developed by:

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Date: July 2025

1. Project Overview

*This is a fully functional full-stack task management web application built independently using the **MERN stack** – MongoDB, Express.js, React.js, and Node.js. The application allows users to **add, view, and manage tasks** through an intuitive interface.*

The project was developed entirely by me, demonstrating my capability to design, develop, and deploy both frontend and backend components without external assistance.

2. Technologies & Tools Used

- **Frontend:** React.js, HTML, CSS
 - **Backend:** Node.js, Express.js
 - **Database:** MongoDB
 - **API Tools:** RESTful API, Axios, Postman
 - **Version Control:** Git, GitHub
 - **IDE:** Visual Studio Code
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3. Project Features

- *Add, display, and manage tasks using a clean and responsive interface*
 - *Frontend communicates with backend via REST APIs*
 - *State managed using React Hooks*
 - *API tested and debugged using Postman*
 - *Persistent data storage using MongoDB*
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4. Project Architecture

- *The **React frontend** interacts with the backend through **Axios** for data handling*
 - *The **Express backend** contains all the logic and route handling*
 - ***MongoDB** is used as a NoSQL database to store task data*
 - *The project follows a modular and scalable code structure*
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5. My Role & Responsibilities

Since this was an individual project, I was responsible for every part of development:

- *Designed and built the **entire backend** with Node.js & Express*
- *Developed the **frontend** interface using React.js*
- *Created and tested all **REST APIs***
- *Integrated the frontend and backend using Axios*
- *Handled **state management** and UI logic*
- *Managed the project with **Git** and deployed locally*

6. Challenges & Solutions

- **Full-stack Integration:** Solved async and CORS issues during API communication
- **Debugging APIs:** Used Postman extensively to test and fix backend routes
- **React State Issues:** Learned proper use of hooks to manage updates efficiently
- **Learning Curve:** Self-taught many tools while building the project, including MongoDB schema design

7. Outcome & Learnings

- Built a complete web app from scratch
- Improved my ability to manage a software project independently
- Strengthened my understanding of REST architecture and component-based UI design
- Gained real-world experience in handling bugs, tools, and version control

8. Conclusion

This project is a strong reflection of my technical abilities and learning mindset. It shows that I can not only write code but also design and build a full product independently. I believe this demonstrates the readiness and potential I can bring to an internship position in software development.

THANK YOU